

LYMPHOID NEOPLASIA

IGH::FENDRR and specific *KRAS* mutations define a novel B-ALL molecular subtype with poor chemotherapy response

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KEY POINTS

- **FOXF1/FENDRR is a molecular B-ALL subtype in adults defined by *IGH::FENDRR*, specific *KRAS* mutations, and *FOXF1* overexpression.**
- **Patients with FOXF1/FENDRR B-ALL show poor initial chemotherapy response salvaged by immunotherapy or hematopoietic stem cell transplantation.**

Large-scale sequencing efforts have defined up to 27 diagnostic subtypes in B-cell precursor acute lymphoblastic leukemia (B-ALL), leaving few samples unclassified. Extended genomic and transcriptomic profiling in routine diagnostics broadens the sample collection, enabling identification of novel subtypes. We analyzed 4857 patients with B-ALL from 3 cohorts and identified a group of 20 patients (age, 18-66 years; median, 34 years) characterized by a previously undescribed *IGH::FENDRR* rearrangement exclusive to this subtype ($n = 17/20$), *KRAS* p.A146T/V/P mutations ($n = 17/20$ vs $n = 86/4857$; $P < .001$), and distinct DNA methylation/gene expression profiles, including overexpression of the lncRNA *FENDRR* and the transcription factor *FOXF1* (*FOXF1/FENDRR*) as well as JAK/STAT and RAS/MAPK signaling signatures. A gene expression machine learning classifier accurately identified *FOXF1/FENDRR* cases in 2 independent cohorts. Patients treated according to German Multicenter Study Group for Adult ALL (GMALL)/Group for Research in Adult ALL (GRAALL) protocols showed very poor chemotherapy response with 8 of 13 exhibiting induction failure or minimal residual disease (MRD) $\geq 10^{-3}$ and 8 of 12 remaining MRD positive after first consolidation/salvage. Intensification including blinatumomab ($n = 10$) and/or allogeneic stem cell transplantation ($n = 12$) resulted in ongoing molecular remission in 13 of 16 patients. *FOXF1/FENDRR* represents a novel B-ALL subtype which might benefit from early immunotherapeutic treatment or targeted interventions.

Introduction

The molecular landscape of B-cell precursor acute lymphoblastic leukemia (B-ALL) is shaped by genomic aberrations and corresponding transcriptional profiles.¹ The International Consensus Classification of Myeloid Neoplasms and Acute Leukemias and the Fifth edition of the World Health Organization Classification of Lymphoid Neoplasms define up to 22 molecular subtypes, as well as 5 provisional entities.^{2,3} However, 10% to 15% of patients cannot be assigned to recognized

subtypes, suggesting the existence of yet unidentified drivers. Here we report a novel molecular subtype defined by a rearrangement of *IGH* to the *FOXF1/FENDRR* locus and particular *KRAS* mutations.

Study design

We analyzed 1715 adult patients with B-ALL treated according to the Multicenter Study Group on Adult ALL (GMALL) protocols with the EuroClonality NGS DNA Capture Assay

Table 1. Clinical characteristics of patients with FOXF1/FENDRR B-ALL

	Patient																			
	1	2	3	4	5	6	7	8	9	10	11	B_SL122	G14_0280	G14_0727	B_SL445	B_SL1005	SJERG 021891	SJALL 054487	SJALL 054499	SJBALL 014249
Cohort	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GMALL	GRAALL	GRAALL	GRAALL	GRAALL	GRAALL	St. Jude	St. Jude	St. Jude	St. Jude
Sex	F	M	M	M	F	M	F	M	M	M	M	F	M	M	F	M	M	F	F	M
Age (y)	46	32	22	41	19	40	66	48	34	33	28	23	30	33	36	19	22	55	37	18
WBC (× 10 ⁹ /L)	2.30	1.70	0.86	1.61	2.60	4.28	1.48	1.23	1.44	4.30	2.10	2.20	1.00	2.38	1.74	2.60	2.70	0.90	1.30	NA
Immuno-phenotype	c-ALL	c-ALL	c-ALL	c-ALL	Pre-B-ALL	c/pre-B-ALL	c-ALL	c/pre-B-ALL	c-ALL	Pre-B-ALL	Pre-B-ALL	c-ALL	c-ALL	c-ALL	Pre-B-ALL	c-ALL	NA	NA	NA	NA
MRD post Ind	4.00E-04	NA	Positive <1E-05	8.00E-04	9.00E-03	Positive <1E-04	—	NA	1.10E-01	7.00E-03	3.00E-02	5.00E-01	3.00E-01	7.00E-01	5.00E-04	2.00E-01	NA	NA	NA	NA
MRD post Cons 1	1.00E-03	7.00E-04	Negative <1E-05	Positive <1E-03	1.00E-02	Negative <3E-05	—	NA	NA	2.00E-02	Negative <1E-05	2.00E-04	2.00E-04	Negative <1E-05	5.00E-04	3.00E-01	NA	NA	NA	NA
Blina	Yes	No	Yes	Yes	Yes	No	—	No	Yes	Yes	No	Yes	No	Yes*	Yes†	Yes‡	NA	NA	NA	NA
Allo-HSCT	Yes	Yes	Yes	Yes	Yes	No	—	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	NA	NA	NA	NA
Outcome	Alive in CR at 2099 d	Alive in CR at 944 d	Alive in CR at 742 d	Alive in CR at 377 d	Alive in CR at 574 d	Alive in CR at 666 d	Died during induction therapy	Alive in CR at 519 d	Alive in CR at 458 d	Alive in CR at 1709 d	Alive in CR at 92 d	Died at 212 d	Alive in CR at 1174 d	Alive in CR at 337 d	Relapsed at 761 d	Alive at 498 d	NA	NA	NA	NA

CAR, chimeric antigen receptor; CR, complete remission; F, female; IG/TR, immunoglobulin/T-cell receptor rearrangements; M, male; NA, not available; VANDA-BLINA, combination of blinatumomab and VANDA (etoposide, cytarabine, mitoxantrone, dexamethasone and asparaginase) regimen; WBC, white blood cells.

*VANDA-BLINA salvage.

†After relapse.

‡After loss of B-cell aplasia after CAR T-cell therapy.

§No MRD measurements due to failure to detect IG/TR markers.

||Received early blinatumomab during induction therapy followed by allo-HSCT due to induction failure.

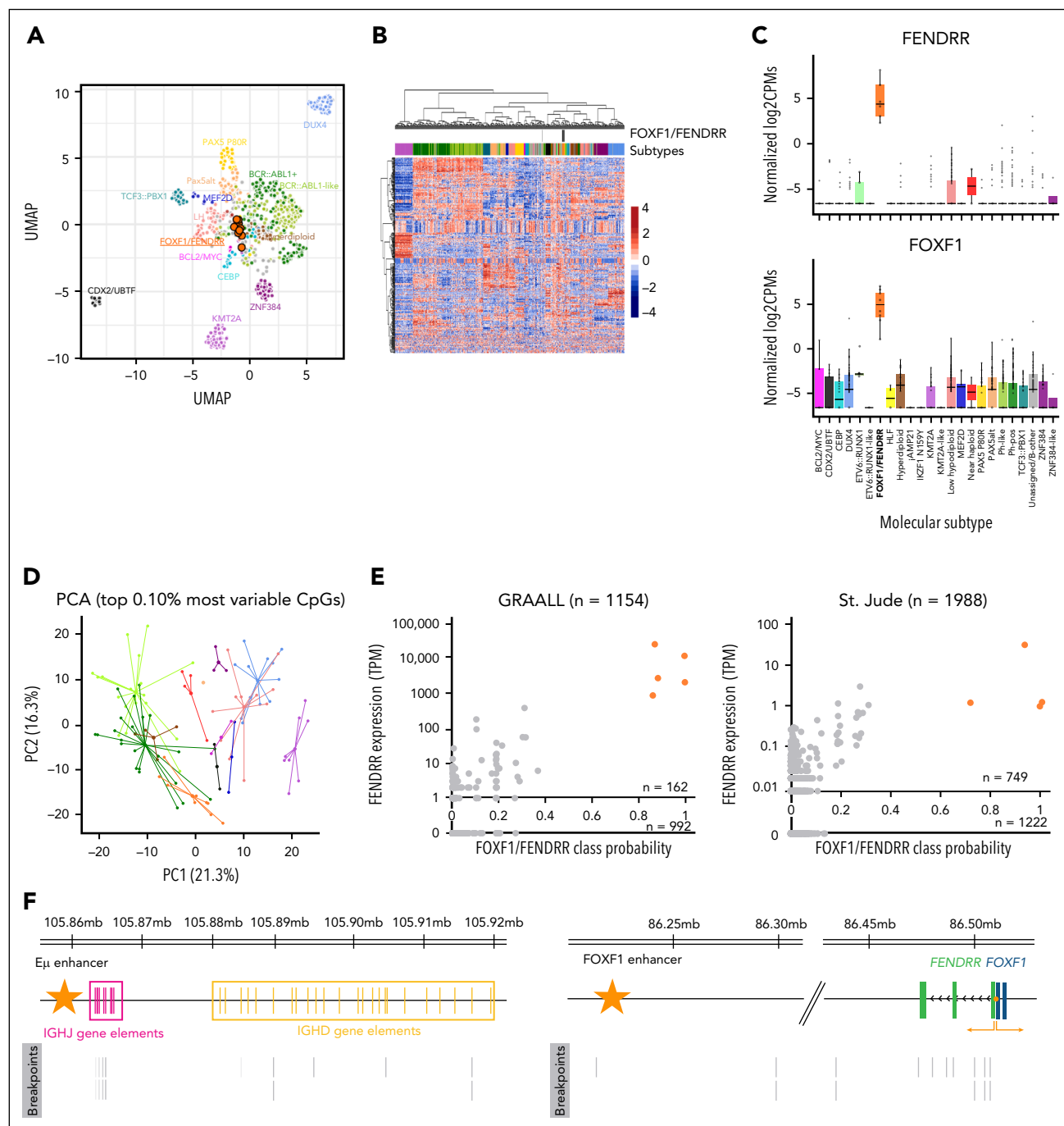


Figure 1. Molecular profiling of a novel B-ALL subtype: FOXF1/FENDRR. (A) UMAP of 767 adult patients with B-ALL with WTS data from the discovery cohort. (B) Unsupervised clustering of the top 300 variably expressed genes of 767 adult patients with B-ALL with WTS data from the discovery cohort. (C) Expression of *FENDRR* and *FOXF1* across molecular subtypes ($n = 767$). (D) Methylation profile of patients of FOXF1/FENDRR subtype ($n = 9/11$) and a B-ALL cohort of other subtypes ($n = 83$).¹ Principal component analysis of the top 0.1% most variable CpGs are shown. Lines connect centroids of the previously established B-ALL subtypes.¹ Subtype colors are consistent with panel A. (E) *FENDRR* expression vs class probability of a machine learning classifier trained on the discovery cohort for the identification of FOXF1/FENDRR cases. Results from 2 independent validation cohorts are shown. *FENDRR* itself was not included in the classifier. (F) Genomic breakpoints of patients with *IGH::FENDRR* rearrangement of the discovery cohort ($n = 11$) on chromosome 14 and chromosome 16. (G) Genomic profiling of patients of FOXF1/FENDRR subtype from the discovery cohort ($n = 11$) and the 2 validation cohorts ($n = 9$). (H) Distribution of *KRAS* mutations in the FOXF1/FENDRR subtype ($n = 20$) compared with other B-ALL subtypes ($n = 3688$). A total of $n = 13$ patients with multiple *KRAS* mutations including codon 146 and other codons, were listed as "other *KRAS* mutations." c-ALL, common ALL; PCA, principal component analysis; TPM, transcripts per million; UMAP, uniform manifold approximation and projection; WBC, white blood cells; WTS, whole transcriptome sequencing.

(EC-NDC Assay, Univ8 Genomics, Belfast, Northern Ireland)⁴ followed by ARResT/Interrogate analysis⁵ ($n = 981$) and/or whole transcriptome sequencing using the Illumina stranded mRNA protocol followed by IntegrateALL and ALLCatchR

analysis^{6,7} ($n = 767$). A machine learning classifier for identification of patients with FOXF1/FENDRR B-ALL was trained on 707 patient samples by performing least absolute shrinkage and selection operator selection⁸ and a 10-fold randomized

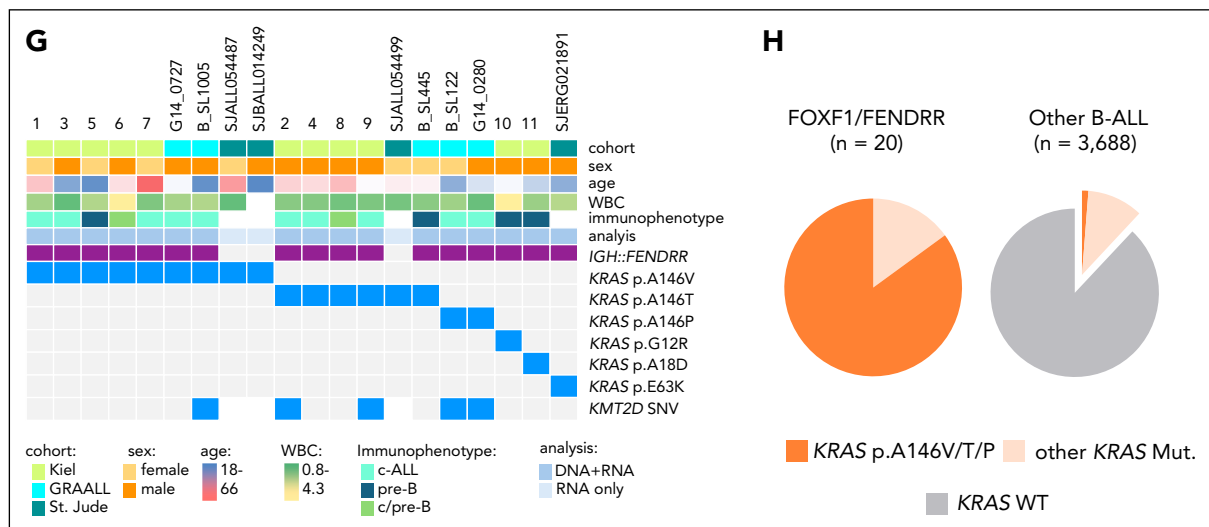


Figure 1 (continued)

stratified cross-validation scheme with an ensemble of 5 machine learning methods implemented in the R package caret.⁹ Subsequently we applied it to 2 independent B-ALL cohorts for identification of additional patients: a cohort of 1154 adult B-ALL samples from the Group for Research in Adult ALL (GRAALL) study group and the St. Jude cohort reported by Gu et al^{10,11} (n = 1988). DNA methylation patterns were profiled for 9 of 20 patients using the Infinium MethylationEPIC version 2.0 array and compared to 83 primary diagnostic B-ALL samples. Copy number alterations were assessed by multiplex ligation-dependent probe amplification using the SALSA multiplex ligation-dependent probe amplification Probemix P335 ALL-IKZF1 (MRC Holland) and single nucleotide polymorphism array-based genotyping using the Illumina Infinium Genotyping System (iScan) according to the manufacturer's instructions. Ex vivo drug response profiling (DRP) was performed for 1 xenograft derived from a patient with FOXF1/FENDRR B-ALL, by exposing leukemic cells to a panel of selected inhibitors and conventional chemotherapeutics for 72 hours as previously described.^{12,13} All patients provided written informed consent in accordance with the Declaration of Helsinki. This study was approved by the institutional review board of the Christian Albrechts University in Kiel, Germany (ethics approval number D416/21). Detailed information on the methods is available in the supplemental Methods, available on the *Blood* website.

Results and discussion

From 4857 patients with B-ALL (3572 adults and 1285 children), we identified 20 (13 male and 7 female) patients with recurrent distinct characteristics, identifying them as a novel molecular subtype. Age at initial diagnosis ranged from 18 to 66 years (median, 34 years) and all presented with a pre-B/common-B-ALL immunophenotype. Compared with an overall GMALL08 cohort (aged 18-55 years), these patients had significantly lower white blood cell count at diagnosis, with a median of $1.74 \times 10^9/L$ (range, $0.86 \times 10^9/L$ to $4.3 \times 10^9/L$) vs $7.7 \times 10^9/L$ (range, $0.3 \times 10^9/L$ to $731.9 \times 10^9/L$; $P < .001$; Table 1).

Unsupervised analysis of the GMALL cohort B-ALL patients (n = 77/767) without confirmed subtype allocation identified a distinct cluster for 9 out of 77 patients (Figure 1A-B) characterized by the absence of recurrent B-ALL subtype-defining driver aberrations and 799 differentially expressed genes (FDR < 0.05, log2FC > 2) compared to the remaining B-ALL samples (supplemental Figure 1). The most strikingly upregulated genes were Forkhead protein 1 (FOXF1) and its adjacent FOXF1 enhancer-derived noncoding RNA (FENDRR; Figure 1C). We therefore termed this group FOXF1/FENDRR. Further included in this signature were the largely unstudied *OLFM3*, the argonaute proteins *PIWIL1* and *PIWIL3*, as well as *KDR* (previously: *VEGFR2*), a critical mediator of VEGF-driven signaling pathways, which is regulated by *FOXF1*.^{14,15} Using our ALLCatchR⁶ tool for B-ALL subtype prediction, most patients showed BCR::ABL1-like candidate predictions, while lacking typical BCR::ABL1-like ALL drivers (n = 15/18). DNA methylation profiling confirmed independence of a FOXF1/FENDRR cluster despite some overlap with BCR::ABL1-positive/-like B-ALL (Figure 1D; supplemental Figure 3). We identified 34 genes as specific to this group by feature selection (supplemental Figure 2A-B) and implemented them in the ALLCatchR 2802-gene signature for B-ALL subtype discrimination.⁶ We trained a machine learning classifier for these FOXF1/FENDRR cases, which specifically identified all 9 of 9 patients with RNA sequencing (RNA-seq) in the training set. Testing on the independent GRAALL and St. Jude cohorts identified 5 of 1154 and 4 of 1988 additional patients with FOXF1/FENDRR B-ALL, respectively (Figure 1E).

In 17 out of these 20 patients and none of the remaining 4837 patients, we detected the so far not described *IGH::FENDRR* rearrangement by genomic (n = 12/12, NDC-Capture panel/whole genome sequencing) and/or transcriptomic (n = 9/18 RNA-seq) profiling (Figure 1F). The lack of fusion transcript detection by RNA-seq in some cases may reflect a low blast percentage in the respective samples (blast percentage, 11%-50%). Genomic *FENDRR* breakpoints were intronic (n = 6/12) or downstream (n = 6/12) of *FENDRR* (supplemental Table 1). *FENDRR* expression was limited to the first exon in cases with intronic breakpoints, whereas downstream breakpoints

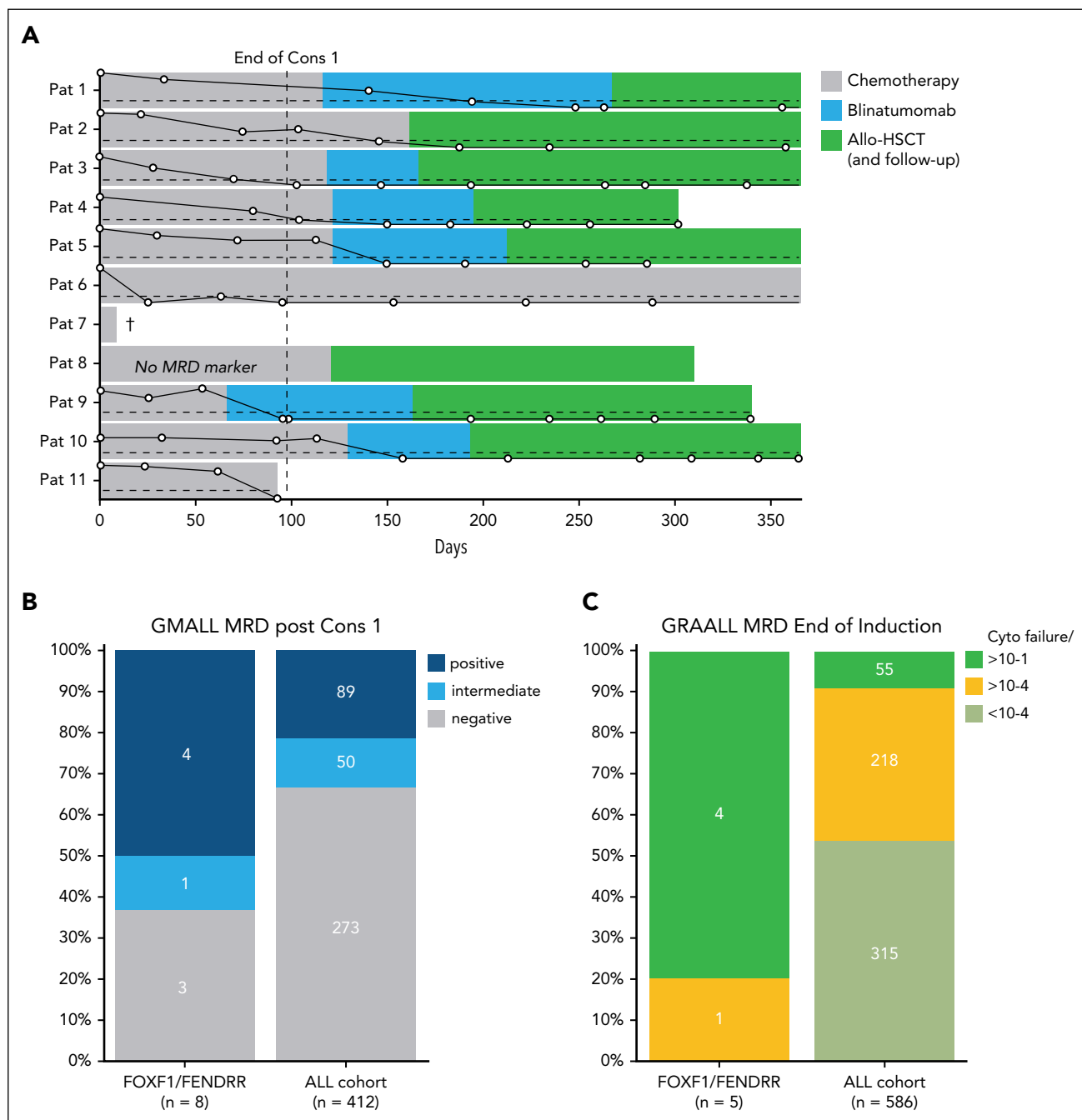


Figure 2. Therapy response of patients of the FOXF1/FENDRR subtype. (A) MRD kinetics of patients of the FOXF1/FENDRR subtype (n = 11) within the first year of treatment. Information on blinatumomab and allogeneic stem cell transplantation (allo-HSCT) is included. Each dot represents an MRD measurement. Dots below the dotted line are MRD positive and quantifiable, and dots on the dotted line represent MRD positive values below the quantitative range. Dots below the line are MRD negative. (B) MRD response of patients of the FOXF1/FENDRR subtype of the discovery cohort treated according to GMALL protocols measured by clone-specific real-time quantitative PCR of immune gene rearrangements after consolidation cycle I in comparison to adult patients with B-ALL of other subtypes (n = 412, GMALL 08 patients with available MRD data). MRD negative: MRD negative with a sensitivity of at least 10⁻⁴; MRD intermediate: MRD positive <10⁻⁴ or below quantifiable range; MRD positive: MRD above 10⁻⁴. Patients 7, 8, and 9 were excluded. (C) MRD response of patients of the FOXF1/FENDRR subtype from the French cohort treated according to GRAALL/FRALLE protocols (n = 5) after the end of induction in comparison to adult patients with B-ALL of other subtypes (n = 586, GRAALL-2014²⁵). Pat, patient; PCR, polymerase chain reaction.

resulted in expression of the complete lncRNA (supplemental Figure 4). *FOXF1* was highly expressed in all cases, consistent with the assumption that the *IGH* enhancer is redirected to drive expression of *FOXF1* and the remaining *FENDRR* exons. During treatment courses, *IGH::FENDRR* levels were comparable to established IG markers, supporting its specificity for

the leukemic clone and its potential applicability as a minimal residual disease (MRD) marker (supplemental Figure 5).

In addition, genomic profiling identified *KRAS* mutations in all patients of the FOXF1/FENDRR subtype, representing the only recurrent secondary aberration (supplemental Table 1). The most

frequent variants were p.A146V/T/P (n = 17/20), whereas p.G12R, p.A18D, and p.E63K were detected in 1 case each (Figure 1G-H). This represented a strong enrichment of codon 146 variants compared with patients of other B-ALL subtypes (85% vs 1%; $P < .001$; Figure 1H). Notably, the overall less frequent codon 146 or p.G12R and p.A18D variants have been shown to induce weaker downstream MAPK activity compared to typical hotspots (eg, KRAS G12D),¹⁶⁻¹⁹ implying a “Goldilocks principle” of MAPK signaling in FOXF1/FENDRR B-ALL. FOXF1 has been shown to regulate STAT3,²⁰ and it is by itself subjected to JAK2-dependent regulation.^{21,22} Consistently, we observed upregulation of both RAS/MAPK and JAK/STAT signaling pathways in FOXF1/FENDRR gene set enrichment analyses (supplemental Figure 6), which might reflect the partial overlap of DNA methylation/gene expression signatures with BCR::ABL1-positive/-like ALL. DRP data on 1 patient-derived xenograft sample (patient 6) functionally supported this biology. Sensitivity to MEK inhibitors (trametinib, selumetinib) indicated KRAS mutation-driven RAS-MAPK pathway activation (supplemental Figure 7), whereas sensitivity to PI3K/AKT inhibitors (buparlisib, ipatasertib) indicated parallel PI3K/AKT pathway activation. Notably, a series of inhibitors against receptor and nonreceptor tyrosine kinases (VEGFR, FLT3, PDGFR, ABL1), including dovitinib and sunitinib, as well as fedratinib and ponatinib, showed distinct effects *ex vivo*, in line with similarities of FOXF1/FENDRR transcriptomes to BCR::ABL1-positive/-like ALL. Conversely, resistance to dexamethasone, cytarabine, and the BCL2 inhibitor venetoclax suggests survival is kinase-driven and BCL2-independent. Collectively, the findings from the DRP assay are consistent with the rearrangement broadly activating downstream signaling pathways, which will merit further functional evaluation.

Patients of the FOXF1/FENDRR subtype and available outcome data were treated according to the GMALL (n = 11; Figure 2A; Table 1), GRAALL (n = 4), or French Acute Lymphoblastic Leukaemia study group (FRALLE; n = 1) protocols. One patient in the GMALL group died during induction. Of the remaining 10, 2 were stratified high risk due to MRD persistence and 8 to standard risk. One patient received early blinatumomab during induction therapy followed by allogeneic stem cell transplantation (allo-HSCT) due to induction failure; in another, MRD measurements failed. Of the remaining 8, only 3 patients (38%) achieved complete MRD clearance after first consolidation chemotherapy indicating a poor chemotherapy response compared with a GMALL reference cohort (66% [273/412] MRD negativity at the same time point [Figure 2B]). Six of these 8 patients subsequently received blinatumomab and/or allo-HSCT. Similarly, 4 of 5 GRAALL/FRALLE patients failed induction therapy with MRD levels >10% (Figure 2C). After first consolidation, only 1 patient cleared MRD. Subsequent treatment included blinatumomab (n = 2), with one of these patients later proceeding to chimeric antigen receptor T-cell therapy (n = 1), and allogeneic stem cell transplantation (n = 4), collectively supporting a poor chemotherapy response phenotype in patients of the FOXF1/FENDRR subtype.

FOXF1/FENDRR has recently been identified as a driver of a novel lineage-ambiguous high-risk T-ALL subtype in which FOXF1/FENDRR expression is driven from retargeting of the BCL11B enhancer.²³ B-ALL samples with other lineage-transcending ALL drivers like KMT2A- or ZNF384-rearrangements show characteristics of more immature B-ALL developmental states.²⁴ Patients with FOXF1/FENDRR B-ALL,

however, showed the highest proximity to more advanced pre-B II stages consistent with the observed common B-ALL immunophenotypes (supplemental Figure 8).

In conclusion, *IGH::FENDRR* and specific *KRAS* mutations define a novel B-ALL subtype with a distinct gene expression/DNA methylation signature and poor chemotherapy response. Co-occurrence of these driver events suggests an oncogenic synergy that not only drives resistance to chemotherapy, but also exposes potential therapeutic vulnerabilities within the MAPK and PI3K/AKT signaling pathways.

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Authorship

Contribution: M.B., L.B., S.B., A.M.H., and W.W. designed the study; S.B., W.W., A.M.H., and B.K. collected and selected patient data; R.K., M.P., Q.G., T.L., N.B., I.I., C.G.M., and E.C. collected and analyzed data for validation cohorts; N.G., T.L., and N.B. provided clinical data; S.B. and W.W. analyzed and interpreted EC-NDC data and MRD tracking; S.B., A.M.H., R.K., M.P., Q.G., E.C., and L.B. analyzed and interpreted transcriptome data; A.M.H., T.B., and N.W. performed integrative bioinformatic data analysis; J.M.H., L.A.S., L.B., A.K., and H.B. analyzed methylation data; K.I. and L.B. analyzed copy number variations; S.B. and A.M.H. performed statistical tests and produced visualizations; F.D.S., L.C., B.B., and J.-P.B. performed and interpreted drug response profiling; I.I., C.G.M., and E.C. contributed to the conceptual framework of the study; M.B., L.B., and C.D.B. supervised the study; S.B. wrote the first draft of the manuscript; and all authors approved the final version of the manuscript.

Conflict-of-interest disclosure: C.D.B. reports consulting for Astellas, Bristol Myers Squibb, AstraZeneca, Amgen, Jazz Pharmaceuticals, Gilead, and Janssen. M.B. is a member of the board of directors or advisory committees for Incyte and Amgen; and serves on the speakers bureau for BD, Janssen, Pfizer, and Amgen. C.G.M. has received speaking fees from Amgen; grant support from Pfizer and AbbVie; and royalties from Cyrus. The remaining authors declare no competing financial interests.

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Footnotes

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RNA sequencing (RNA-seq) count data for all FOXF1/FENDRR cases and the remaining B-ALL cohort are publicly available at <https://doi.org/10.5281/zenodo.18310354> and <https://doi.org/10.5281/zenodo.18952693>. Raw sequencing data for RNA-seq and gene panel sequencing are available in part through the European

Genome-Phenome Archive under accession numbers EGAD00001008633 and EGAD00001010070.

Original data are available from the author, Sonja Bendig (sonja.bendig@uksh.de).

The online version of this article contains a data supplement.

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